FACULTY OF COMPUTER APPLICATIONS

Marwadi University
Marwadi Chandarana Group

NAAC A+

**Faculty of Computer Applications**

# BCA Sem-1
# (AY-2026-27)

# Lab Book

## Sub Code : 05BC3103
## (Programming Practices : 1)

| | |
|---|---|
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| **Full Name :** | Vivek kumar |

**GitHub Link |** https://github.com/Vivek -606/BCA-1-PP-1--92600527127 - Vivek Kumar

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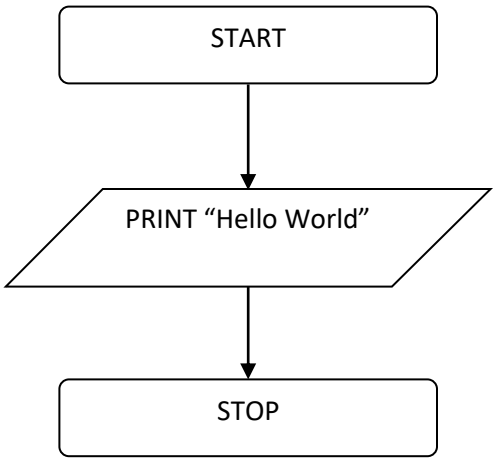

| Unit : 1 | |
|---|---|
| **Program No** | **Definition** |
| **1.** | Print "Hello World". |
| **2.** | 2 Values from user and Perform All Arithmetic Operations. |
| **3.** | Print numbers from 1 to 20. |
| **4.** | Input 3 values from user and find out its Average. |
| **5.** | Calculate the Square and Cube of a number. |
| **6.** | Interchange the value of two numbers. |
| **7.** | Find largest out of three numbers. |
| **8.** | Find out Fibonacci series up to given n values. |
| **9.** | Find out Number is Odd or Even. |
| **10.** | Find out sum of digits of a given number. |

# (1) Print "Hello World".

## Algorithm :

Step 1 : START
Step 2 : PRINT “Hello World”
Step 3 : STOP

## Flowchart :



**Code :**

```
#include<stdio.h>
#include<conio.h>

void main(){
        clrscr();
        printf("Hello World");
        getch();
}
```

**Output :**

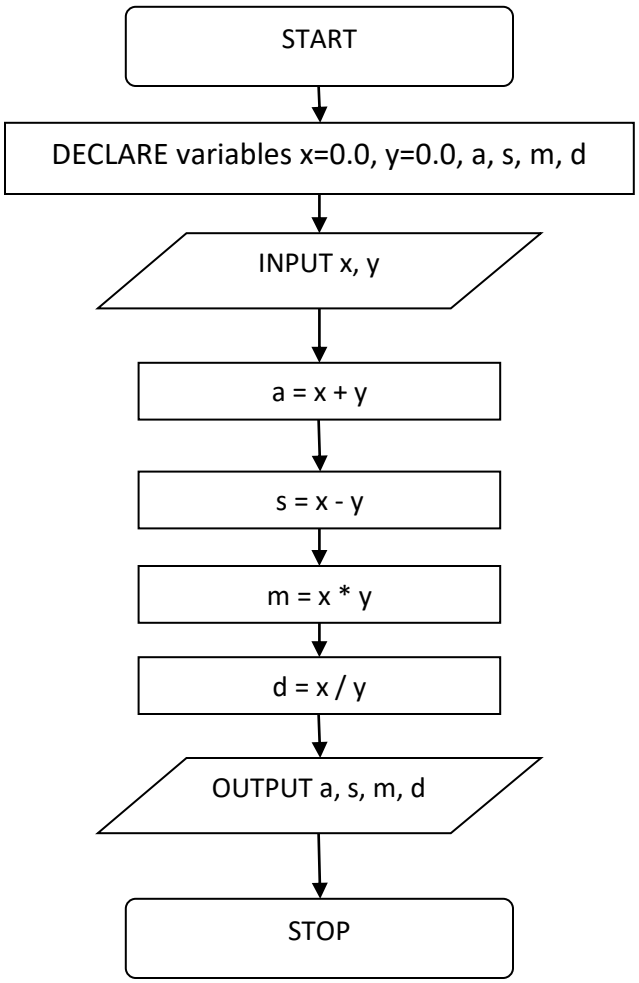


## (2) 2 Values from user and Perform All Arithmetic Operations.

### Algorithm :

Step 1 : START
Step 2 : DECLARE variables x=0.0, y=0.0, a, s, m, d
Step 3 : INPUT x,y
Step 4 : a = x + y
Step 5 : s = x – y
Step 6 : m = x * y
Step 7 : d = x / y
Step 8 : OUTPUT a, s, m, d
Step 9 : STOP

### Flowchart :



**Code :**

```
#include<stdio.h>
#include<conio.h>

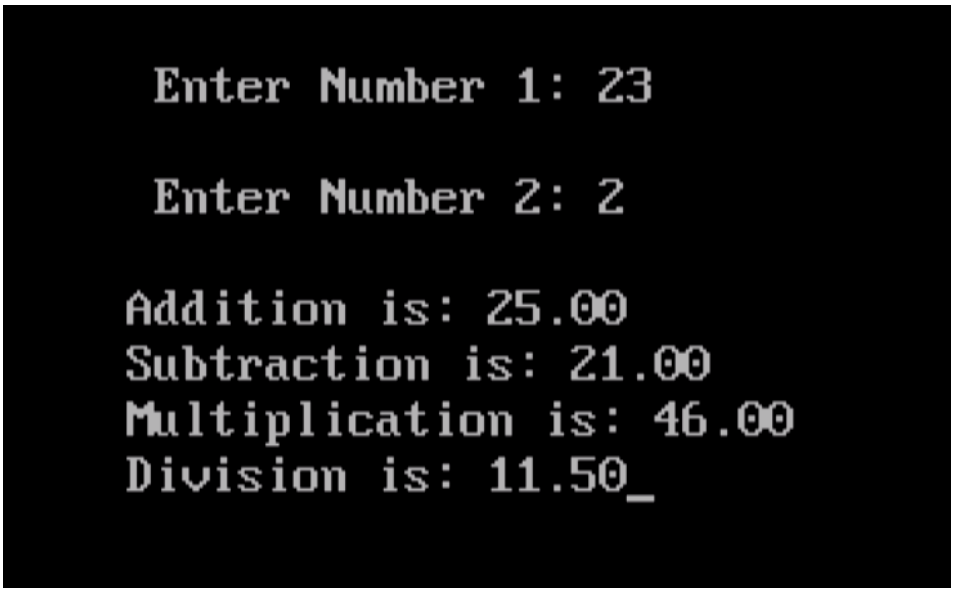
void main()
{
                float x,y;
                clrscr();
                printf("\n Enter Number 1: ");
                scanf("%f",&x);
                printf("\n Enter Number 2: ");
                scanf("%f",&y);

                printf("Addition Is: %.2f \n",x+y);
                printf("Subtraction Is: %.2f \n",x-y);
                printf("Multiplication Is: %.2f \n",x*y);
                printf("Division Is: %.2f \n",x/y);

                getch();

}
```

**Output :**

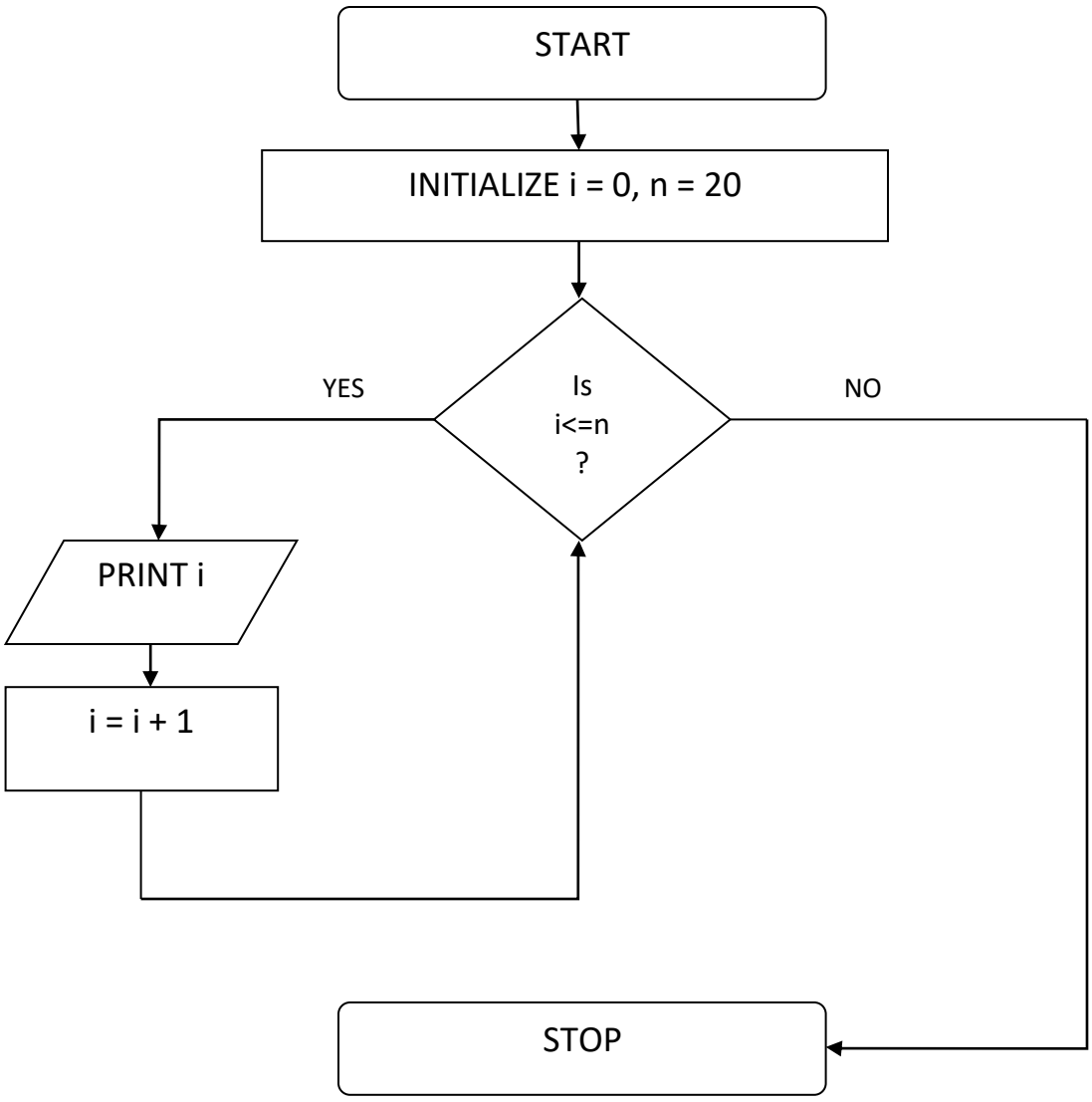


## (3) Print numbers from 1 to 20.

### Algorithm :

Step 1 : START
Step 2 : INITIALIZE i = 0, n = 20
Step 3 : REPEAT STEP WHILE i<=n
PRINT i
i = i + 1
GOTO Step 3
Step 4 : STOP

### Flowchart :



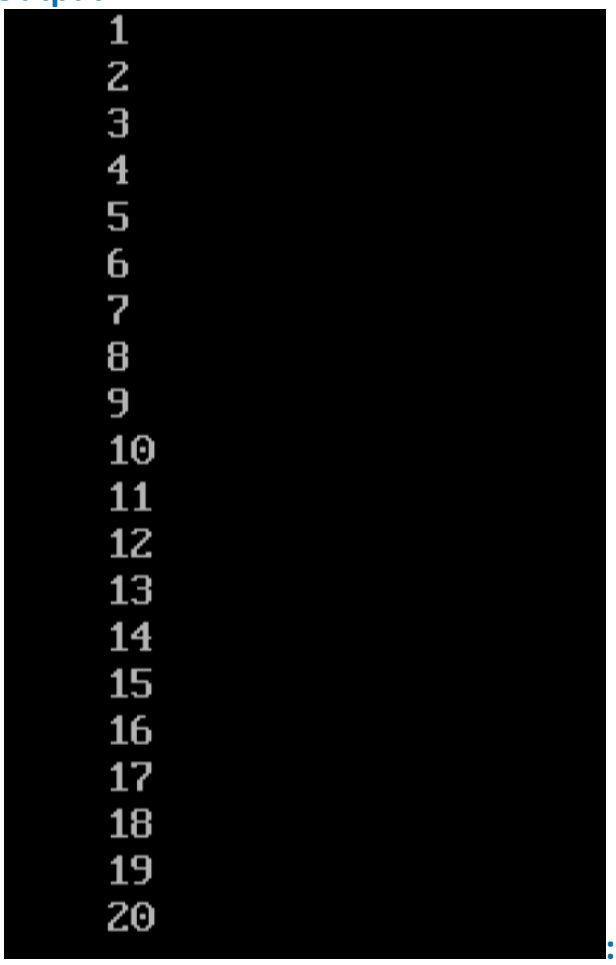
**Code :**

```
#include<stdio.h>
#include<conio.h>

void main(){
                int i;
                clrscr();
                for (i=1;i<=20;i++){
                  printf("%d\n",i);
                }
                getch();

}
```

**Output**

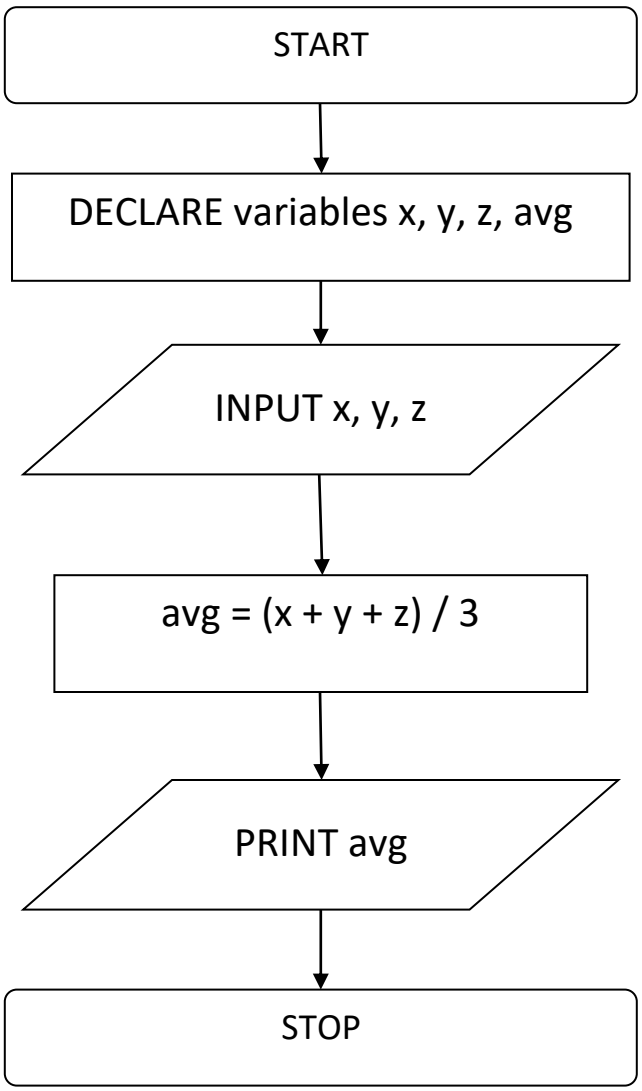


## (4) Input 3 values from user and find out its Average.

### Algorithm :

Step 1 : START
Step 2 : DECLARE variables x, y, z, avg
Step 3 : INPUT x, y, z
Step 4 : avg = (x + y + z) / 3
Step 5 : PRINT avg
Step 6 : STOP

### Flowchart :



**Code :**

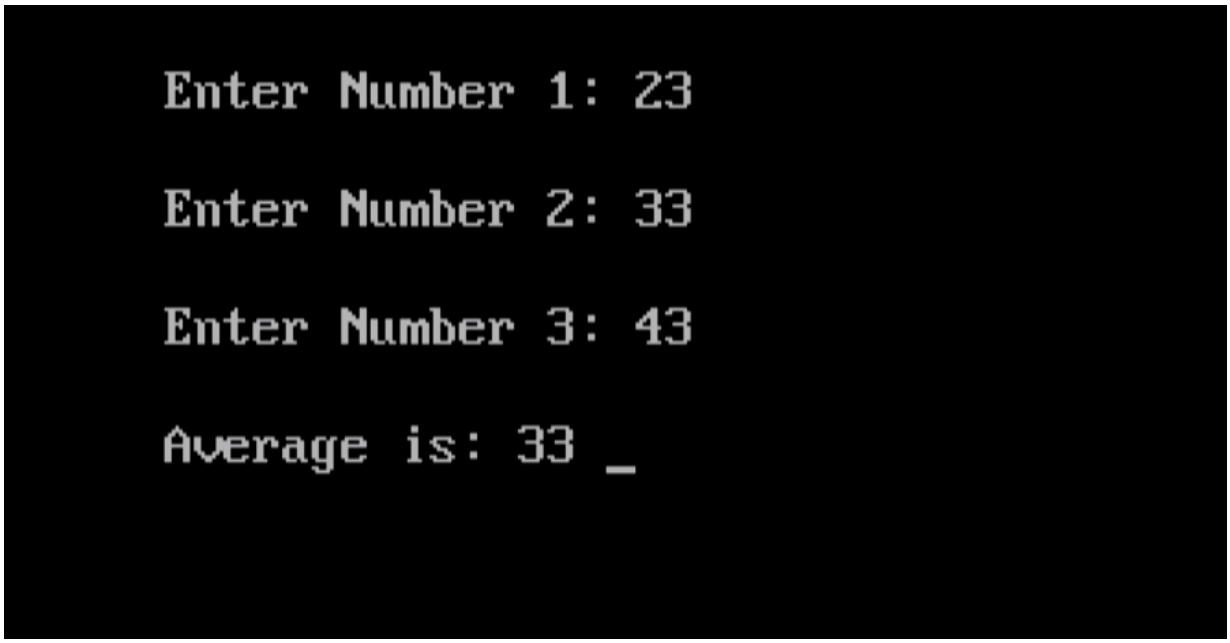
```
#include<stdio.h>
#include<conio.h>

void main()
{
int x,y,z;
clrscr();
printf("\n Enter Number 1: ");
scanf("%d",&x);
printf("\n Enter Number 2: ");
scanf("%d",&y);
 printf("\n Enter Number 3: ");
scanf("%d",&z);
printf("\n Average is: %d ",(x+y+z)/3);
getch();




}
```

**Output :**

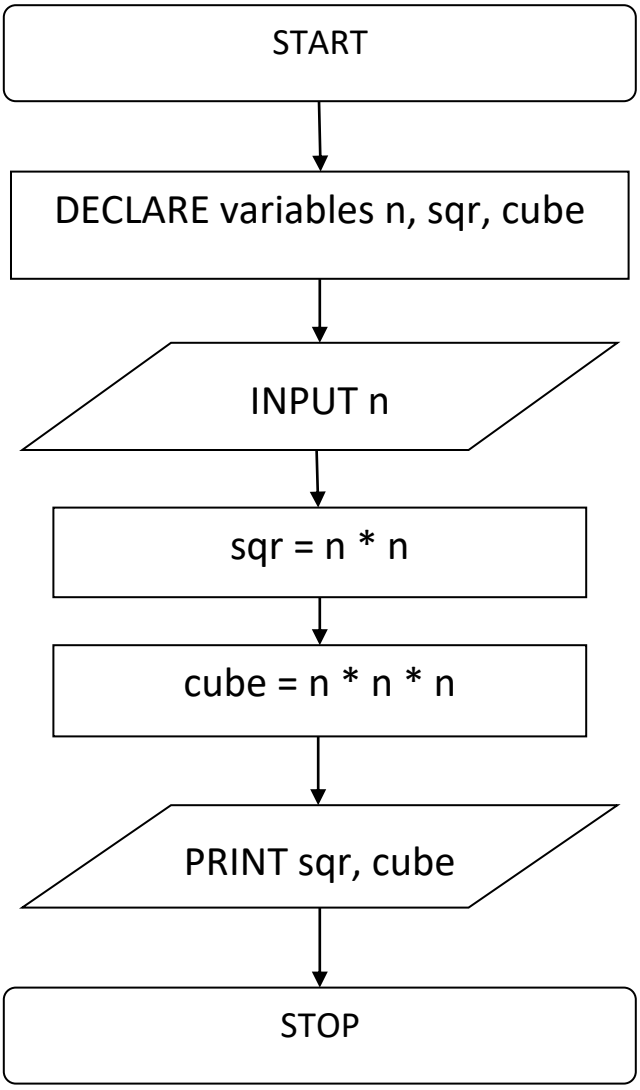


## (5) Calculate the Square and Cube of a number.

**Algorithm :**

Step 1 : START
Step 2 : DECLARE variables n, sqr, cube
Step 3 : INPUT n
Step 4 : sqr = n * n
Step 5 : cube = n* n* n
Step 6 : OUTPUT sqr, cube
Step 7 : STOP

**Flowchart :**



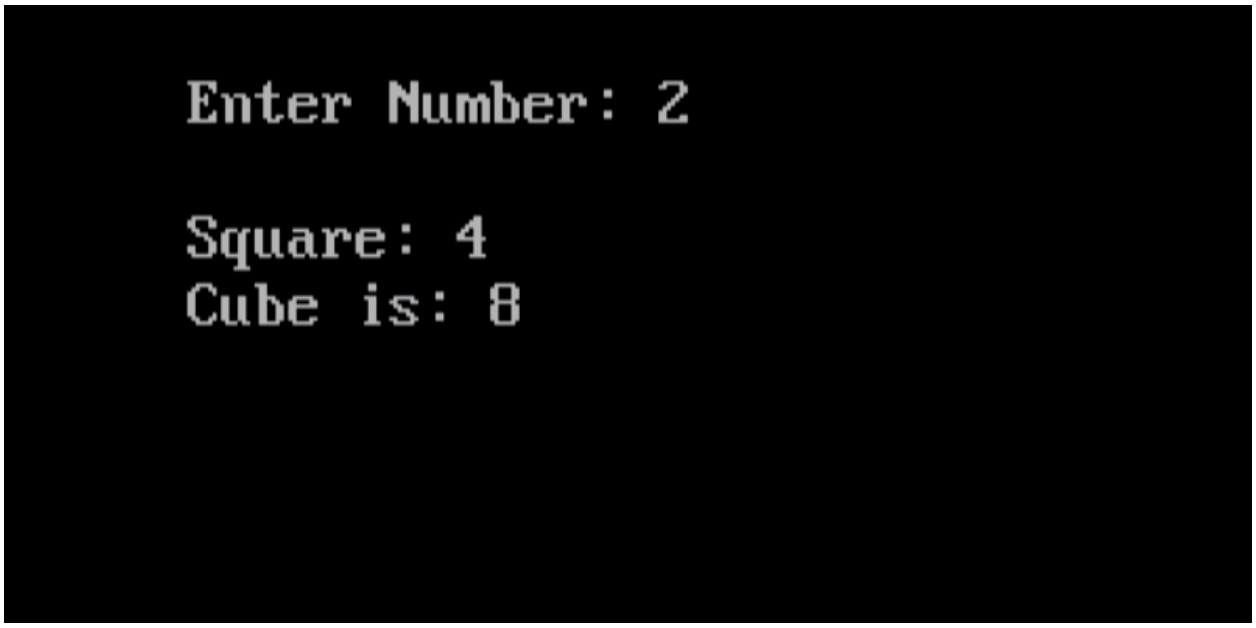
**Code :**

```
#include<stdio.h>
#include<conio.h>

void main()
{
                int n;
                clrscr();
                printf("\n Enter Number: ");
                scanf("%d",&n);
                printf("\n Enter Square: %d",n*n);
                printf("\n Cube is: %d",n*n*n);
                getch();

}
```

**Output :**

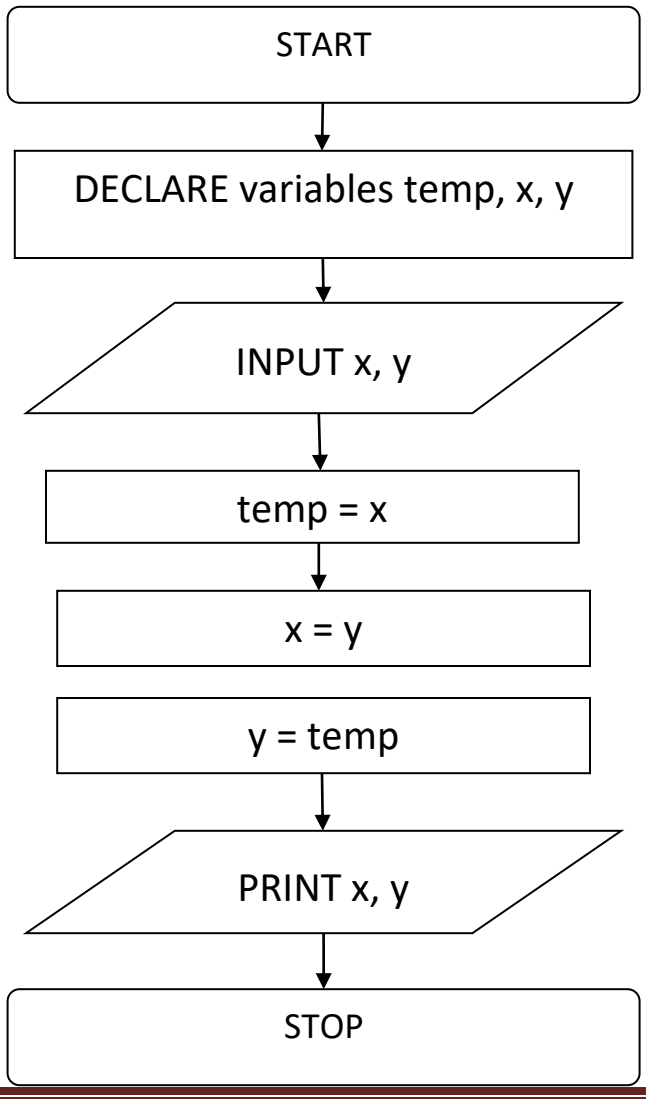


## (6) Interchange the value of two numbers.

### Algorithm :

Step 1 : START
Step 2 : DECLARE variables temp, x, y
Step 3 : INPUT x, y
Step 4 : temp = x
Step 5 : x = y
Step 6 : y = temp
Step 7 : PRINT x, y
Step 8 : STOP

### Flowchart :



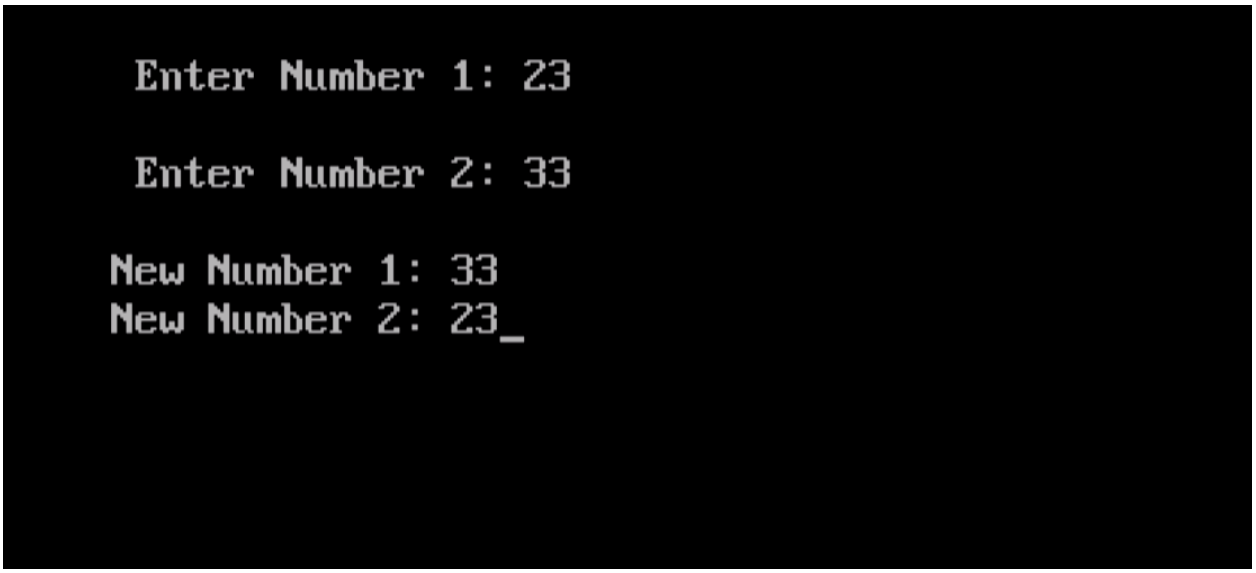
**Code :**

```
#include<stdio.h>
#include<conio.h>

void main()
{
                int temp,x,y;
                clrscr();
                printf("\n Enter Number 1: ");
                scanf("%d",&x);
                printf("\n Enter Number 2: ");
                scanf("%d",&y);
                temp=x;
                x=y;
                y=temp;
                printf("\nNew Number 1: %d",x);
                printf("\nNew Number 2: %d",y);
                getch();

}
```

**Output :**

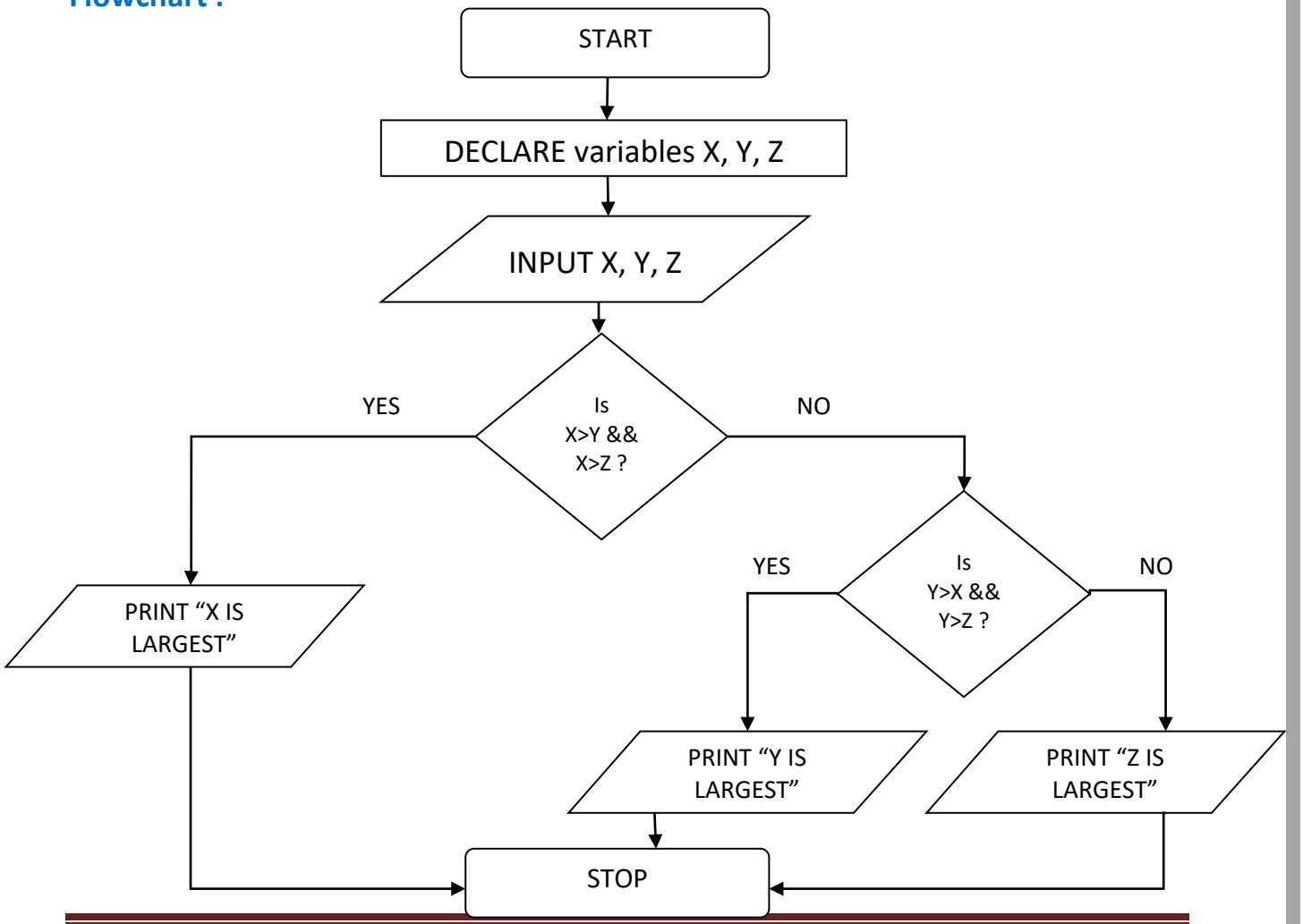


## (7) Find largest out of three numbers.

### Algorithm :

Step 1 : START
Step 2 : DECLARE variables X, Y, Z
Step 3 : INPUT X, Y, Z
Step 4 : IF X > Y && X > Z THEN
PRINT “X is Largest”
Step 5 : ELSE IF Y > Z && Y > X THEN
PRINT “Y is Largest”
Step 6 : ELSE
PRINT “Z is Largest”
Step 7 : STOP

### Flowchart :



**Code :**

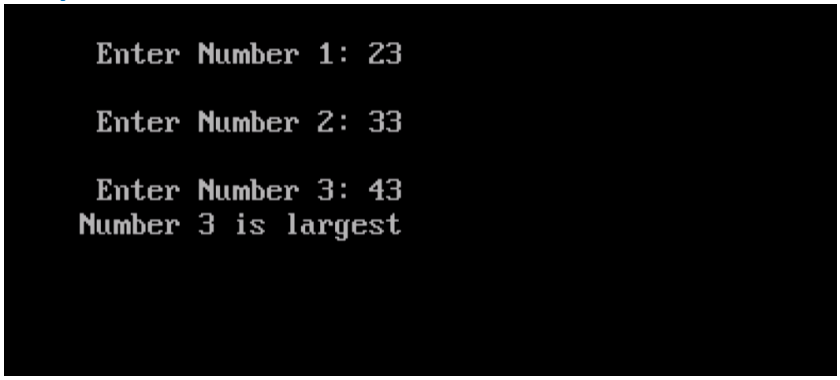
```
#include<stdio.h>
#include<conio.h>

void main(){
                int x,y,z;
                clrscr();
                printf("\n Enter Number 1: ");
                scanf("%d",&x);
                printf("\n Enter Number 2: ");
                scanf("%d",&y);
                printf("\n Enter Number 3: ");
                scanf("%d",&z);

                if(x>y && x>z){
                        printf("Number 1 is largest");
                }
                else if(y>z && y>x){
                        printf("Number 2 is largest");
                }
                else{
                        printf("Number 3 is largest");
                }
                getch();

}
```

**Output :**

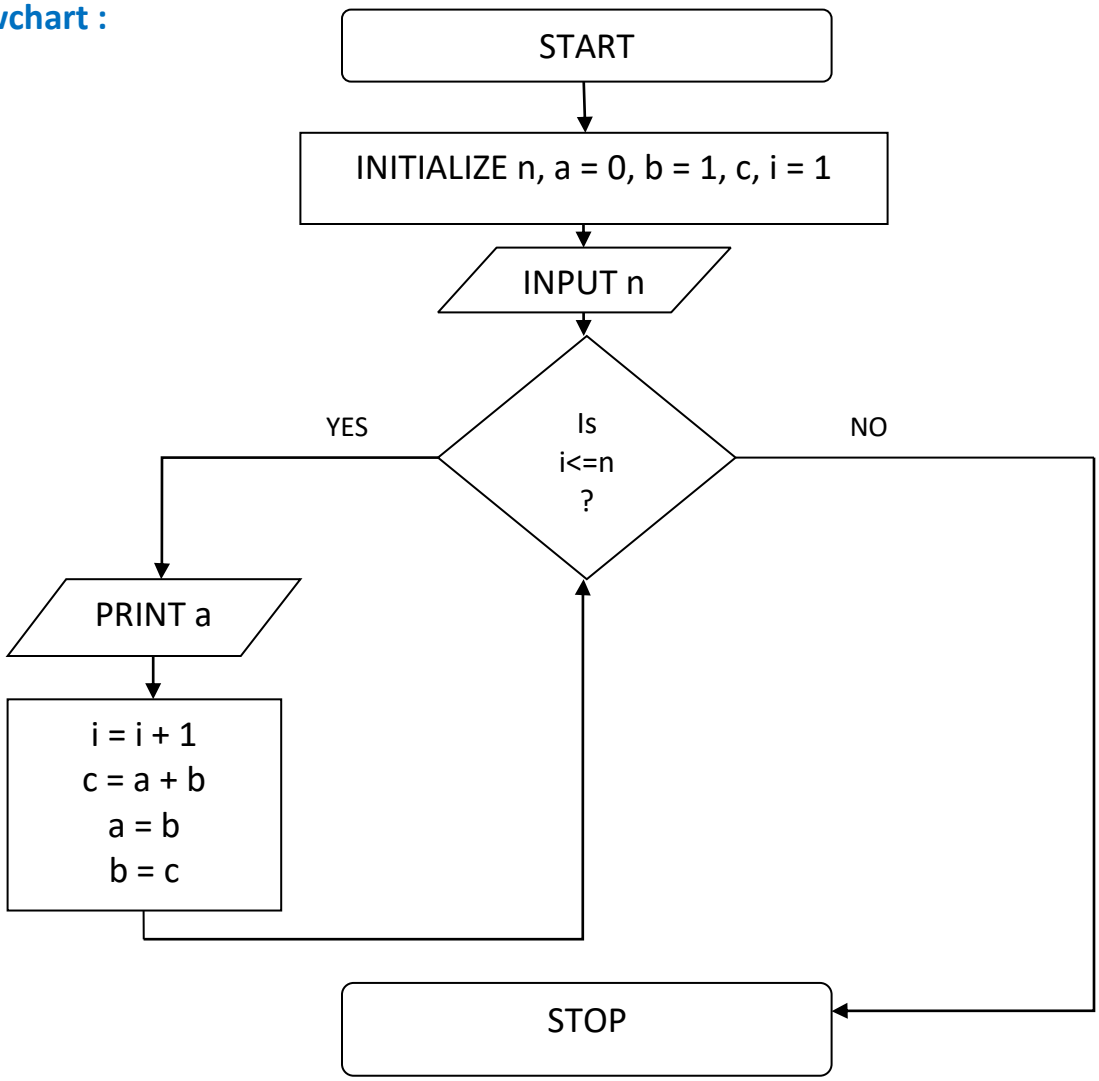


## (8) Find out Fibonacci series up to given n values.

**Algorithm :**

Step 1 : START
Step 2 : INITIALIZE n, a = 0, b = 1, c, i = 1
Step 3 : REPEAT STEP WHILE i<=n
PRINT a
c = a + b
a = b
b = c
i = i + 1
GOTO Step 3
Step 4 : STOP

**Flowchart :**

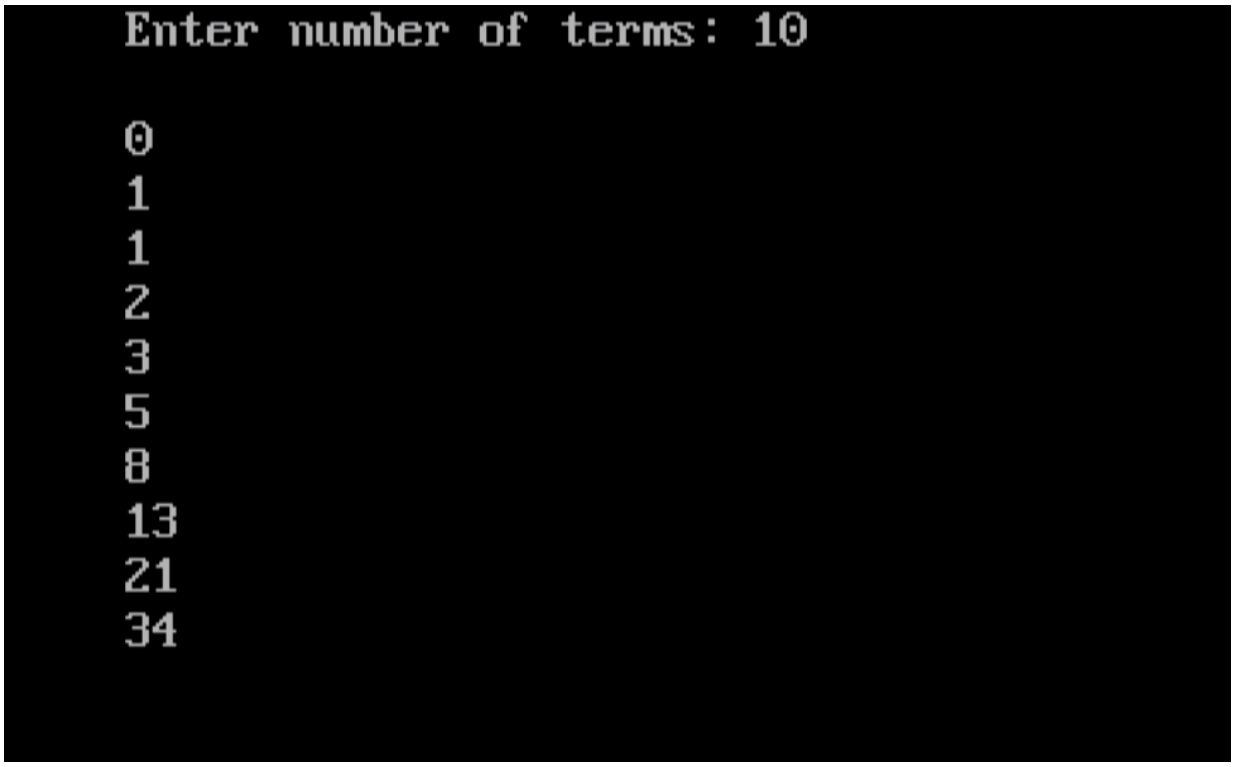


**Code :**

```
#include<stdio.h>
#include<conio.h>

void main(){
                int n, a = 0, b = 1, c,i;
                clrscr();
                printf("Enter number of terms: ");
                scanf("%d", &n);
                for(i = 1; i <= n; i++){
                      printf("%d ", a);
                      c = a + b;
                      a = b;
                      b = c;
                }
                getch();
}
```

**Output :**

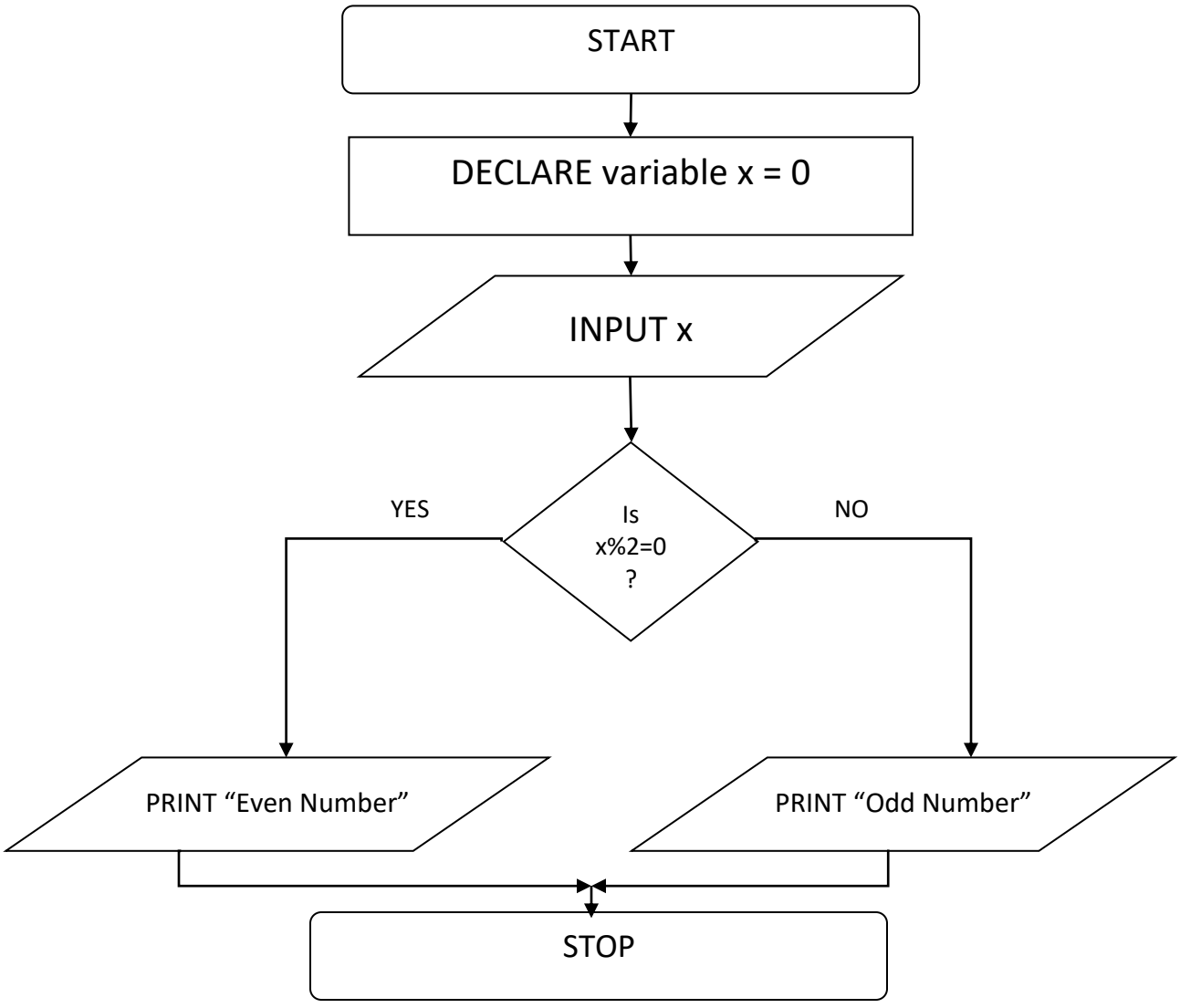


## (9) Find out Number is Odd or Even.

**Algorithm :**

Step 1 : START
Step 2 : DECLARE variable x = 0
Step 3 : INPUT x
Step 4 : IF x % 2 = 0 THEN
PRINT “Even Number”
Step 5 : ELSE
PRINT “Odd Number”
Step 6 : STOP

**Flowchart :**



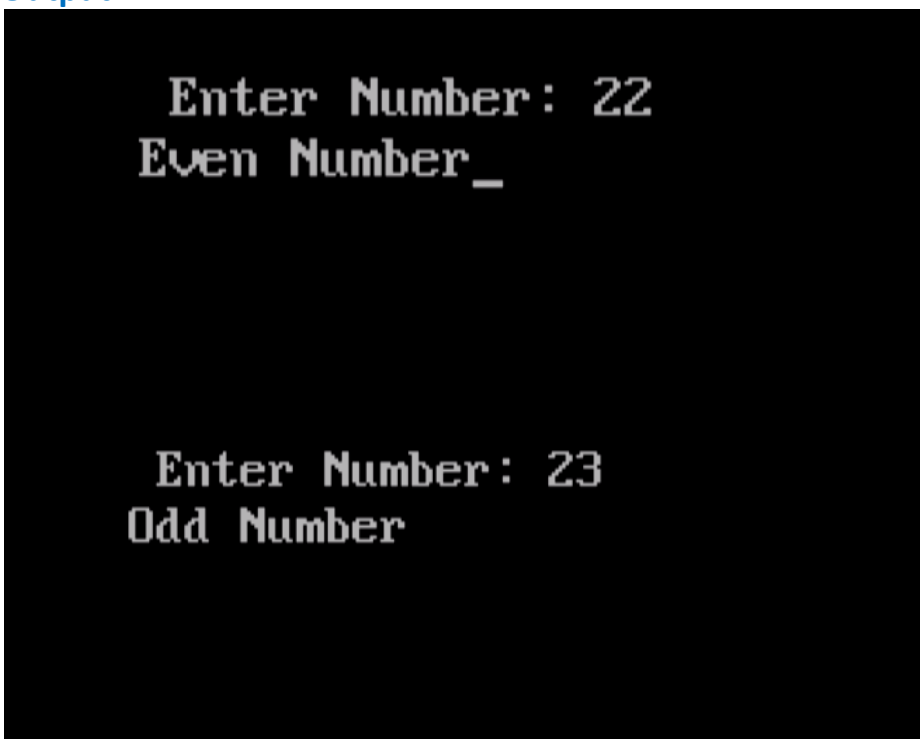
**Code :**

```
#include<stdio.h>
#include<conio.h>

void main()
{
                int n;
                clrscr();
                printf("\n Enter Number: ");
                scanf("%d",&n);

                if(n%2==0){
                    printf("Even Number");
                }
                else
                    printf("Odd Number");
                getch();
}
```

**Output :**

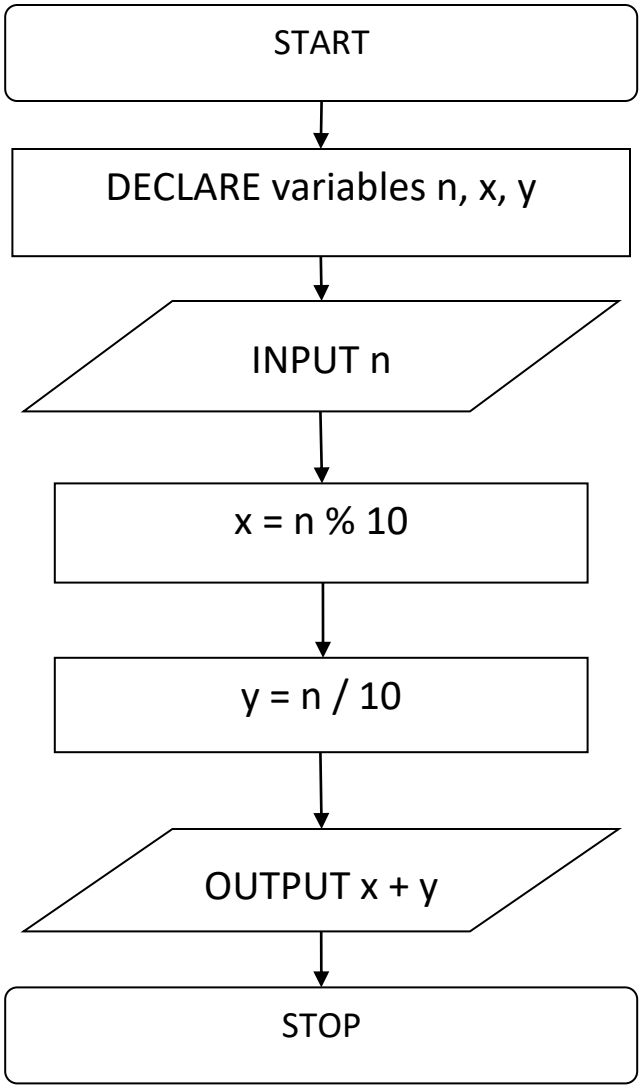


## (10) Find out sum of digits of a given number.

### Algorithm :

Step 1 : START
Step 2 : DECLARE variables n, x, y
Step 3 : INPUT n
Step 4 : x = n % 10
Step 5 : y = n / 10
Step 6 : OUTPUT x + y
Step 7 : STOP

### Flowchart :



**Code :**

```
#include<stdio.h>
#include<conio.h>

void main()
{
   int n,x,y;
   clrscr();
   printf("\n Enter Two Digit Number: ");
   scanf("%d",&n);
   x=n%10;
   y=n/10;
   printf("Sum Of Digits is: %d",x+y);
   getch();

}
```

**Output :**

